ALEENA VARGHESE

AI ENGINEER | GENERATIVE AI | RAG & AI WORKFLOWS

Malappuram, Kerala, India | aleenavar29@gmail.com | +91 6238310759

LinkedIn: linkedin.com/in/aleena-varghese2904 | GitHub: github.com/aleenavarghese29 | Portfolio

PROFESSIONAL SUMMARY

Computer Science (Data Analytics) postgraduate with practical experience building LLM-powered applications, RAG pipelines, multimodal retrieval systems, and AI automation workflows. Experienced with Python, LangChain, LightRAG, vector databases, embeddings, REST APIs, document processing, and prompt engineering. Currently working as an Associate AI Developer Intern, contributing to knowledge-base preparation, chunking, RAG-related tasks, and workflow automation. Interested in developing reliable AI agents with tool use, memory, and grounded access to organisational knowledge.

TECHNICAL SKILLS

Programming & Data: Python, SQL, JavaScript, Pandas, NumPy
LLMs & Generative AI: LLM API integration, prompt engineering, RAG, embeddings, semantic search, context-aware question answering
AI Frameworks & Retrieval: LangChain, LightRAG, ChromaDB, vector retrieval, knowledge graphs, document chunking
AI Applications & APIs: FastAPI, Django, REST APIs, Streamlit, PydanticAI
Machine Learning: Scikit-learn, SVM, TensorFlow/Keras, deep learning, computer vision, NLP
Data & Developer Tools: PostgreSQL, SQLite, Git, GitHub, Docker, MinerU, OpenAI CLIP, VLM concepts

PROFESSIONAL EXPERIENCE

Associate AI Developer Intern | *August 2026 – Present*
LamaxAI Technologies Pvt. Ltd. | Remote, Kerala

- Contribute to AI application and workflow development, including knowledge-base preparation and RAG-related tasks.
- Prepare and review Markdown knowledge-base content and support chunking and content validation for retrieval-based chatbots.
- Learn and work with n8n-based workflow automation, including webhook-triggered workflows and integrations between AI components.
- Support technical documentation, testing, and review of chatbot behavior and knowledge retrieval quality.

AI Engineer Intern | *March 2026 – June 2026*
Lightrains Technolabs | Remote, Thiruvananthapuram, Kerala

- Contributed to a multimodal RAG assistant for technical manuals, combining document processing, retrieval, and LLM-based question answering.
- Worked with LightRAG and knowledge-graph retrieval concepts to connect technical entities, procedures, and relationships.
- Used MinerU-based document processing concepts for parsing documents, extracting layout and tables, and preparing content for retrieval.
- Explored multimodal retrieval using Vision Language Models (VLMs) for visual grounding and semantic retrieval of figures alongside text.
- Supported retrieval/context design, API integration, testing, and technical documentation for the AI application.

SELECTED AI PROJECTS

LLM-Powered AI Gmail Assistant

Python, LangChain, Google Gemini, ChromaDB, Sentence Transformers, Streamlit

- Built a RAG-based assistant to answer natural-language questions over Gmail content.
- Implemented semantic retrieval using embeddings and ChromaDB, with email summarisation, categorisation, and conversational question answering.
- Integrated Gmail API, retrieval components, and Streamlit into an interactive application.

HR Assistant System

Django, React, TypeScript, Llama 3.3, SQLite

- Developed an AI-powered assistant for employee questions about payroll, leave management, company policies, and HR documentation.

- Built role-aware responses and integrated secure REST APIs, role-based access control, and audit logging.

SignBot – Sign Language Recognition

Python, OpenAI CLIP, SVM, Google Gemini AI, Streamlit

- Built a sign-language recognition pipeline using CLIP feature extraction and an SVM classifier.
- Integrated Gemini-generated responses and text-to-speech to support accessible text and voice communication.

Age, Gender, and Emotion Prediction

Python, TensorFlow, Keras, OpenCV

- Developed CNN-based models for age, gender, and facial-emotion prediction using UTKFace and FER2013 datasets.
- Implemented preprocessing, model training, evaluation, and inference workflows.

EDUCATION

M.Sc. Computer Science (Data Analytics)

Digital University Kerala, Thiruvananthapuram | CGPA: 9.12 | 2024–2026

B.Sc. Mathematics

Mar Ivanios College (Autonomous), Thiruvananthapuram | CGPA: 7.71 | 2021–2024

CERTIFICATIONS

- Docker Foundations Professional Certificate — LinkedIn Learning
- Introduction to Large Language Models — Google Cloud
- Google Cloud Career Launchpad — Data Analytics & Computing Foundations
- Python Data Structures — University of Michigan (Coursera)
- Introduction to Git and GitHub — Google (Coursera)
- Data Analytics Job Simulation — Deloitte
- Data Visualization — Tata Group

PROFESSIONAL SKILLS

Analytical thinking • Problem solving • Technical documentation • Teamwork • Adaptability • Continuous learning